TritonNav: A Campus Navigation App for UCSD First-Year and Transfer Students

San Diego Undergraduate Tech Conference — Track 2: Emerging Project

Diego [Last Name] • UC San Diego

Motivation

Starting college at UC San Diego is exhilarating — and disorienting. As a first-year student, I experienced firsthand the anxiety of not knowing where I was going: arriving late to class because I couldn't find the right building entrance, missing advising appointments because the office wasn't where I expected, and spending precious minutes every day just figuring out where to be next. Transfers face an even steeper curve, arriving mid-year with no campus orientation and a full course load.

Google Maps exists, but it stops at the building front door. The UCSD website lists room numbers but provides no route. No existing tool combines a student's personal class schedule with step-by-step, room-level directions and campus resource discovery in one place. TritonNav is built to fill that gap.

Project Description

TritonNav is a mobile-first web application designed specifically for incoming UCSD students. After logging in via TritonLink SSO (the same credentials used for WebReg and Canvas), students see their enrolled courses displayed as cards with a photo of each building and a single "GO!" button. Tapping GO initiates turn-by-turn directions from the student's current location to the exact classroom — not just the building.

The app's core features include:

- Schedule-aware navigation: Automatically surfaces your next class based on your enrolled courses, eliminating the need to search for building names.
- Room-level directions: Step-by-step instructions that guide students through building entrances and hallways to the specific room number.
- Campus resource integration: Contextual prompts for dining halls, food distribution events, Triton balances, and nutritional information — surfaced at the right time and place.
- UCSD SSO authentication: Secure TritonLink login so no new account is required.
- General location search: A freeform search bar for finding any campus location beyond scheduled classes.

The current state of the project includes a fully designed Figma prototype (Version 22) with working screens for SSO login, the My Classes home page, navigation view with step-by-step directions, and the dining/resource hub. A functional React + Node.js web application is under active development, with the navigation routing currently prototyped. The next major milestone is integration of the Google Maps API to power real-time, live directions.

Technical Approach & Tools

Layer	Tools & Rationale
Frontend	React.js — component-based UI enabling rapid iteration on the mobile-first interface
Backend	Node.js + Express — lightweight server handling schedule data, user sessions, and API proxying
Authentication	TritonLink SSO (SAML/OAuth) — leverages existing UCSD student credentials
Navigation	Google Maps Platform API — Directions API for real routing, Maps JavaScript API for map rendering, Places API for campus building data
Design	Figma Make — high-fidelity prototype used for user testing and development reference
Deployment	Vercel / Netlify for frontend; Render or Railway for Node.js backend

Development Timeline

Period	Milestone
Now – May 10	Submit application; finalize Figma prototype screens; document proof-of-concept architecture
May – June	Integrate Google Maps Directions API; connect routing to real campus building coordinates; replace mocked navigation with live directions
June – July 15(Checkpoint 1)	Complete working navigation for 30+ UCSD buildings and lecture halls; implement TritonLink SSO with real session management; begin informal user testing with current students
July 15 – Aug 12(Checkpoint 2)	Conduct structured user testing with incoming first-years and transfers; iterate on UX based on feedback; add dining API integration for live balance and menu data
Aug 12 – Sept 4	Polish UI; finalize feature set; write IEEE-format conference paper; prepare presentation or demo
Sept 4+	Submit final paper and presentation; present at SDUTC Fall Quarter conference

Proof of Concept & Feasibility

TritonNav’s feasibility is demonstrated through three existing components. First, a high-fidelity Figma prototype (Version 22) shows the complete user flow from SSO login through class-based navigation to dining integration — validating the UX design before engineering resources are committed. Second, a working React + Node.js codebase is under active development, with the app shell, routing structure, and page components functional. Third, the core technical dependencies — Google Maps Platform and TritonLink SSO — are both well-documented, widely adopted APIs with established integration patterns, making the implementation path low-risk.

The problem is real and personally validated: as a UCSD student who got lost navigating campus as a first-year, I built this tool to solve a pain point I lived. Every incoming cohort of thousands of first-years and transfer students represents a community that would immediately benefit from TritonNav on day one of classes.

Word count: ~420 words • Contact: [your email]